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CS-158A

21 September 2026

Executive Summary: PA1

Bug Narratives:

Q1:

For the `serve_file()` bug, the browser returned a 404 error, which meant that it could not find the requested file. To fix this, I stripped the `'/'` character from the `full_path` variable assignment, resulting in the line being `full_path = os.path.join(WEBROOT, path.lstrip("/"))`. This fix resulted in both `about.html` and `index.html` being served properly to the client.

```
cs158a_pa1: bash
cpetri@afterlife:~/Downloads/cs158a_pa1$ curl -v localhost:8080/index.html
* Host localhost:8080 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
*   Trying [::1]:8080...
* connect to ::1 port 8080 from ::1 port 45526 failed: Connection refused
*   Trying 127.0.0.1:8080...
* Established connection to localhost (127.0.0.1 port 8080) from 127.0.0.1 port 51966
* using HTTP/1.x
> GET /index.html HTTP/1.1
> Host: localhost:8080
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
* Empty reply from server
* shutting down connection #0
curl: (52) Empty reply from server
cpetri@afterlife:~/Downloads/cs158a_pa1$
```

Before `serve_file()` fix

```
cs158a_pa1: bash
cpetri@afterlife:~/Downloads/cs158a_pa1$ curl -v localhost:8080/index.html
* Host localhost:8080 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
*   Trying [::1]:8080...
* connect to ::1 port 8080 from ::1 port 41590 failed: Connection refused
*   Trying 127.0.0.1:8080...
* Established connection to localhost (127.0.0.1 port 8080) from 127.0.0.1 port 59696
* using HTTP/1.x
> GET /index.html HTTP/1.1
> Host: localhost:8080
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Content-Length: 6
< Date: Tue, 22 Sep 2026 04:17:03 GMT
< Server: PA1Server/1.0
< Content-Type: text/html
< Connection: keep-alive
<
* Excess found writing body: excess = 406, size = 6, maxdownload = 6, bytecount = 6
* shutting down connection #0
<!DOCTYPEcpetri@afterlife:~/Downloads/cs158a_pa1$
```

After serve_file() fix

Q2:

For the `build_http_response()` bug, the incorrect amount of content was being transmitted. This was due to the `Content-Length` header being set to `str(len('200 OK'))`. I changed this to be `str(len(body_bytes))` to allow it to transmit the entire body. This was confirmed when the entire `about.html` and `index.html` were served to the client.

```
cs158a_pa1: bash
cpetri@afterlife:~/Downloads/cs158a_pa1$ curl -v localhost:8080/index.html
* Host localhost:8080 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
* Trying [::1]:8080...
* connect to ::1 port 8080 from ::1 port 41590 failed: Connection refused
* Trying 127.0.0.1:8080...
* Established connection to localhost (127.0.0.1 port 8080) from 127.0.0.1 port 59696
* using HTTP/1.x
> GET /index.html HTTP/1.1
> Host: localhost:8080
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Content-Length: 6
< Date: Tue, 22 Sep 2026 04:17:03 GMT
< Server: PA1Server/1.0
< Content-Type: text/html
< Connection: keep-alive
<
* Excess found writing body: excess = 406, size = 6, maxdownload = 6, bytecount = 6
* shutting down connection #0
<!DOCTYPEcpetri@afterlife:~/Downloads/cs158a_pa1$
```

Before build_http_response() bug fix (content length is only 6)

```
cs158a_pa1: bash
cpetri@afterlife:~/Downloads/cs158a_pa1$ curl -v localhost:8080/index.html
* Host localhost:8080 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
* Trying [::1]:8080...
* connect to ::1 port 8080 from ::1 port 37060 failed: Connection refused
* Trying 127.0.0.1:8080...
* Established connection to localhost (127.0.0.1 port 8080) from 127.0.0.1 port 50866
* using HTTP/1.x
> GET /index.html HTTP/1.1
> Host: localhost:8080
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Content-Length: 412
< Date: Tue, 22 Sep 2026 04:19:54 GMT
< Server: PA1Server/1.0
< Content-Type: text/html
< Connection: keep-alive
<
<!DOCTYPE html>
<html>
<head>
  <title>PA1 Index</title>
</head>
<body>
  <h1>PA1 Index Page</h1>
  <p>If you can see this page in your browser, your HTTP server is correctly
    parsing the request, finding this file, and building a valid response.</p>
  <p>Try <a href="/about.html">about.html</a> next, then
    <a href="/missing.html">missing.html</a> to check your 404 handling.</p>
</body>
</html>
* Connection #0 to host localhost:8080 left intact
cpetri@afterlife:~/Downloads/cs158a_pa1$
```

After build_http_request() fix. Note the Content-Length of 412

Part A Recap: Q3

The server has the option to handle a single-request, non-persistent connection or a multi-request, persistent connection. In both cases, the connection handler requests a specific file using `serve_file()`. If the file is found, for example if `index.html` is requested, `serve_file()` returns a tuple `(200, file_content, content_type)`. If the file is not found, for example if `missing.html` is requested, `serve_file()` returns `(404, None, None)`. The connection handler function then calls `builds_http_response()` using the output of `serve_file()` and transmits the built https request.

Part B Recap:

Q4:

The persistent connection exits on 3 conditions. If the client requests a close, we handle it gracefully, sending the final packet with a confirmation of socket closure. If the `parse_http_request()` function returns nothing, then we respond with a 400 Bad Request message and close the connection. If the request isn't a GET request, the server responds with a 405 Method Not Allowed message and closes the connection.

Q5:

If the client doesn't close the connection or sends any data, the socket will time out and close the connection. My code uses the `socket.timeout()` method to handle the timeout implementation. I used the provided `TIMEOUT` global constant which is 10 seconds.

Part C Evidence:

```
PA1:python
(venv) cpetri@afterlife:~/School/networking/assignments/PA1$ python server.py single-threaded
[Server] Running in single-threaded (no concurrency - one connection at a time) mode
[Server] Serving files from: /home/cpetri/School/networking/assignments/PA1/www
[Server] Listening on http://127.0.0.1:8888
[Server] Press Ctrl+C to stop.

[21:27:00.127] [Server] Waiting for the next connection...
[21:27:02.740] [Server] Accepted connection from ('127.0.0.1', 47626)
[21:27:12.766] [Server] Finished handling ('127.0.0.1', 47626)
[21:27:12.766] [Server] Waiting for the next connection...
[21:27:12.766] [Server] Accepted connection from ('127.0.0.1', 47630)
[21:27:12.767] [Server] Finished handling ('127.0.0.1', 47630)
[21:27:12.767] [Server] Waiting for the next connection...
[]

PA1:python
(venv) cpetri@afterlife:~/School/networking/assignments/PA1$ python client.py slow
[SlowClient] Requesting /large.bin from 127.0.0.1:8888, reading slowly on purpose...
21:27:00.754 - Received 16 bytes
21:27:04.254 - Received 32 bytes
21:27:05.754 - Received 48 bytes
21:27:07.255 - Received 64 bytes
21:27:08.755 - Received 80 bytes
21:27:10.255 - Received 96 bytes
21:27:11.755 - Received 112 bytes
21:27:13.256 - Received 128 bytes
21:27:14.756 - Received 144 bytes
21:27:16.256 - Received 160 bytes
[]

PA1:python
(venv) cpetri@afterlife:~/School/networking/assignments/PA1$ python client.py fast
[FastClient] Requesting /about.html from 127.0.0.1:8888...
Finished in 8.162631034851074
(venv) cpetri@afterlife:~/School/networking/assignments/PA1$
```

Single-threaded test complete in 8.16 seconds

```
PA1:python
(venv) cpetri@afterlife:~/School/networking/assignments/PA1$ python server.py
[Server] Running in persistent (HTTP/1.1 keep-alive, multi-threaded) mode
[Server] Serving files from: /home/cpetri/School/networking/assignments/PA1/www
[Server] Listening on http://127.0.0.1:8888
[Server] Press Ctrl+C to stop.

[21:31:52.151] [Server] Waiting for the next connection...
[21:31:57.194] [Server] Accepted connection from ('127.0.0.1', 44710)
[21:31:57.194] [Server] Waiting for the next connection...
[21:31:58.981] [Server] Accepted connection from ('127.0.0.1', 44714)
[21:31:58.982] [Server] Waiting for the next connection...
[21:31:58.983] [Server] Finished handling ('127.0.0.1', 44714)
[]

PA1:python
(venv) cpetri@afterlife:~/School/networking/assignments/PA1$ python client.py slow
[SlowClient] Requesting /large.bin from 127.0.0.1:8888, reading slowly on purpose...
21:31:57.199 - Received 16 bytes
21:31:58.699 - Received 32 bytes
21:32:00.199 - Received 48 bytes
21:32:01.700 - Received 64 bytes
[]

PA1:python
(venv) cpetri@afterlife:~/School/networking/assignments/PA1$ python client.py fast
[FastClient] Requesting /about.html from 127.0.0.1:8888...
Finished in 0.0014979839324951172
(venv) cpetri@afterlife:~/School/networking/assignments/PA1$
```

Multi-threaded test complete in 0.001 seconds

Wireshark Evidence:

The image displays a Wireshark packet capture interface. The top pane shows a list of network packets. Packet 673 is highlighted, showing an HTTP GET request to port 8080. The middle pane shows the details of the selected packet, including the Ethernet II header, Internet Protocol Version 4 header, and the Hypertext Transfer Protocol (HTTP) section. The HTTP section shows a 200 OK response with a content length of 412 bytes and a content type of text/html. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
666	7.173867625	::1	::1	TCP	94	34614 → 8080 [SYN] Seq=0 Win=65476 Len=0 MSS=65476 SACK_
667	7.173895407	::1	::1	TCP	74	8080 → 34614 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
668	7.174050598	127.0.0.1	127.0.0.1	TCP	74	51086 → 8080 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_
669	7.174072379	127.0.0.1	127.0.0.1	TCP	74	8080 → 51086 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=
670	7.174086215	127.0.0.1	127.0.0.1	TCP	66	51086 → 8080 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=123
671	7.174150766	127.0.0.1	127.0.0.1	HTTP	154	GET /index.html HTTP/1.1
672	7.174159713	127.0.0.1	127.0.0.1	TCP	66	8080 → 51086 [ACK] Seq=1 Ack=89 Win=65536 Len=0 TSval=27
673	7.175027821	127.0.0.1	127.0.0.1	HTTP	627	HTTP/1.1 200 OK (text/html)
674	7.175078906	127.0.0.1	127.0.0.1	TCP	66	51086 → 8080 [ACK] Seq=89 Ack=562 Win=65536 Len=0 TSval=
675	7.175394979	127.0.0.1	127.0.0.1	TCP	66	51086 → 8080 [FIN, ACK] Seq=89 Ack=562 Win=65536 Len=0 T
676	7.175493654	127.0.0.1	127.0.0.1	HTTP	192	HTTP/1.1 400 Bad Request
677	7.175534180	127.0.0.1	127.0.0.1	TCP	54	51086 → 8080 [RST] Seq=90 Win=0 Len=0

Frame 673: Packet, 627 bytes on wire (5016 bits), 627 bytes captured on interface 0

Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 8080, Dst Port: 51086, Seq: 1, Win: 0, Len: 0

Hypertext Transfer Protocol

- HTTP/1.1 200 OK\r\n
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
- Content-Length: 412\r\n
 - [Content length: 412]
- Date: Tue, 22 Sep 2026 05:08:24 GMT\r\n
- Server: PAIServer/1.0\r\n
- Content-Type: text/html\r\n
- Connection: keep-alive\r\n
- \r\n
- [Request in frame: 671]
- [Time since request: 877.055 microseconds]
- [Request URI: /index.html]
- [Full request URI: http://localhost:8080/index.html]
- File Data: 412 bytes

Line-based text data: text/html (13 lines)

A successful GET request with code 200, length 412, and typ text/html

tcp.port == 8080

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	:::1	:::1	TCP	94	44998 → 8080 [SYN] Seq=0 Win=65476 Len=0 MSS=65476 SACK=0
2	0.000050655	:::1	:::1	TCP	74	8080 → 44998 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
3	0.000247535	127.0.0.1	127.0.0.1	TCP	74	46776 → 8080 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK=0
4	0.000280276	127.0.0.1	127.0.0.1	TCP	74	8080 → 46776 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65483
5	0.000303950	127.0.0.1	127.0.0.1	TCP	66	46776 → 8080 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=284
6	0.000380103	127.0.0.1	127.0.0.1	HTTP	156	GET /missing.html HTTP/1.1
7	0.000395903	127.0.0.1	127.0.0.1	TCP	66	8080 → 46776 [ACK] Seq=1 Ack=91 Win=65536 Len=0 TSval=30
8	0.001338220	127.0.0.1	127.0.0.1	HTTP	220	HTTP/1.1 404 Not Found
9	0.001360572	127.0.0.1	127.0.0.1	TCP	66	46776 → 8080 [ACK] Seq=91 Ack=155 Win=65536 Len=0 TSval=
10	0.001548595	127.0.0.1	127.0.0.1	TCP	66	46776 → 8080 [FIN, ACK] Seq=91 Ack=155 Win=65536 Len=0 T
11	0.001640337	127.0.0.1	127.0.0.1	HTTP	192	HTTP/1.1 400 Bad Request
12	0.001669442	127.0.0.1	127.0.0.1	TCP	54	46776 → 8080 [RST] Seq=92 Win=0 Len=0

Frame 8: Packet, 220 bytes on wire (1760 bits), 220 bytes captured on interface 0

Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 8080, Dst Port: 46776, Seq: 91, Win: 65536, Len: 0

Hypertext Transfer Protocol

HTTP/1.1 404 Not Found\r\n

Response Version: HTTP/1.1

Status Code: 404

[Status Code Description: Not Found]

Response Phrase: Not Found

Content-Length: 0\r\n

[Content length: 0]

Date: Tue, 22 Sep 2026 05:24:02 GMT\r\n

Server: PA1Server/1.0\r\n

Content-Type: text/html\r\n

Connection: keep-alive\r\n

\r\n

[Request in frame: 6]

[Time since request: 958.117 microseconds]

[Request URI: /missing.html]

[Full request URI: http://localhost:8080/missing.html]

An error 404 for no file found, content length of 0, and type text/html

tcp.port == 8080						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	50542 → 8080 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK=0
2	0.000041878	127.0.0.1	127.0.0.1	TCP	74	8080 → 50542 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495
3	0.000064691	127.0.0.1	127.0.0.1	TCP	66	50542 → 8080 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=338
4	0.000136516	127.0.0.1	127.0.0.1	HTTP	154	GET /index.html HTTP/1.1
5	0.000147917	127.0.0.1	127.0.0.1	TCP	66	8080 → 50542 [ACK] Seq=1 Ack=89 Win=65536 Len=0 TSval=37
6	0.001318012	127.0.0.1	127.0.0.1	HTTP	627	HTTP/1.1 200 OK (text/html)
7	0.001381922	127.0.0.1	127.0.0.1	TCP	66	50542 → 8080 [ACK] Seq=89 Ack=562 Win=65536 Len=0 TSval=
8	0.001807952	127.0.0.1	127.0.0.1	HTTP	154	GET /about.html HTTP/1.1
9	0.002017264	127.0.0.1	127.0.0.1	HTTP	599	HTTP/1.1 200 OK (text/html)
10	0.002163969	127.0.0.1	127.0.0.1	TCP	66	50542 → 8080 [FIN, ACK] Seq=177 Ack=1095 Win=65536 Len=0
11	0.002248338	127.0.0.1	127.0.0.1	HTTP	192	HTTP/1.1 400 Bad Request
12	0.002286479	127.0.0.1	127.0.0.1	TCP	54	50542 → 8080 [RST] Seq=178 Win=0 Len=0

▶ Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (1) on interface 0
▶ Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00
▶ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
▶ Transmission Control Protocol, Src Port: 50542, Dst Port: 8080, Seq: 0, Win: 65495, Len: 0

Persistent server results in only 1 TCP handshake for the 2 sequential requests

tcp.port == 8080						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	49384 → 8080 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_
2	0.000056947	127.0.0.1	127.0.0.1	TCP	74	8080 → 49384 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=
3	0.000086713	127.0.0.1	127.0.0.1	TCP	66	49384 → 8080 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=331
4	0.000208201	127.0.0.1	127.0.0.1	HTTP	154	GET /index.html HTTP/1.1
5	0.000222027	127.0.0.1	127.0.0.1	TCP	66	8080 → 49384 [ACK] Seq=1 Ack=89 Win=65536 Len=0 TSval=39
6	0.001485117	127.0.0.1	127.0.0.1	HTTP	603	HTTP/1.1 200 OK (text/html)
7	0.001504483	127.0.0.1	127.0.0.1	TCP	66	49384 → 8080 [ACK] Seq=89 Ack=538 Win=65536 Len=0 TSval=
8	0.001526214	127.0.0.1	127.0.0.1	TCP	66	8080 → 49384 [FIN, ACK] Seq=538 Ack=89 Win=65536 Len=0 T
9	0.001782304	127.0.0.1	127.0.0.1	TCP	66	49384 → 8080 [FIN, ACK] Seq=89 Ack=539 Win=65536 Len=0 T
10	0.001815697	127.0.0.1	127.0.0.1	TCP	66	8080 → 49384 [ACK] Seq=539 Ack=90 Win=65536 Len=0 TSval=
11	0.001916997	127.0.0.1	127.0.0.1	TCP	74	49390 → 8080 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_
12	0.001938307	127.0.0.1	127.0.0.1	TCP	74	8080 → 49390 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=
13	0.001950971	127.0.0.1	127.0.0.1	TCP	66	49390 → 8080 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=280
14	0.001995154	127.0.0.1	127.0.0.1	HTTP	154	GET /about.html HTTP/1.1
15	0.002000443	127.0.0.1	127.0.0.1	TCP	66	8080 → 49390 [ACK] Seq=1 Ack=89 Win=65536 Len=0 TSval=24
16	0.002732958	127.0.0.1	127.0.0.1	HTTP	575	HTTP/1.1 200 OK (text/html)
17	0.002784765	127.0.0.1	127.0.0.1	TCP	66	49390 → 8080 [ACK] Seq=89 Ack=510 Win=65536 Len=0 TSval=
18	0.002824189	127.0.0.1	127.0.0.1	TCP	66	8080 → 49390 [FIN, ACK] Seq=510 Ack=89 Win=65536 Len=0 T
19	0.003098744	127.0.0.1	127.0.0.1	TCP	66	49390 → 8080 [FIN, ACK] Seq=89 Ack=511 Win=65536 Len=0 T
20	0.003145862	127.0.0.1	127.0.0.1	TCP	66	8080 → 49390 [ACK] Seq=511 Ack=90 Win=65536 Len=0 TSval=

Non-persistent mode results in 2 handshakes, one for each request

